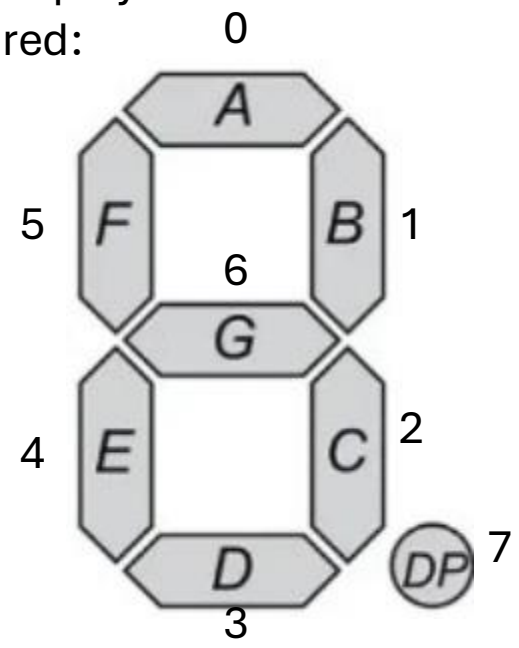
7-Segment Decoder:

Determines the number in LSB of the decimal number

How 7-segment display LED's are numbered:



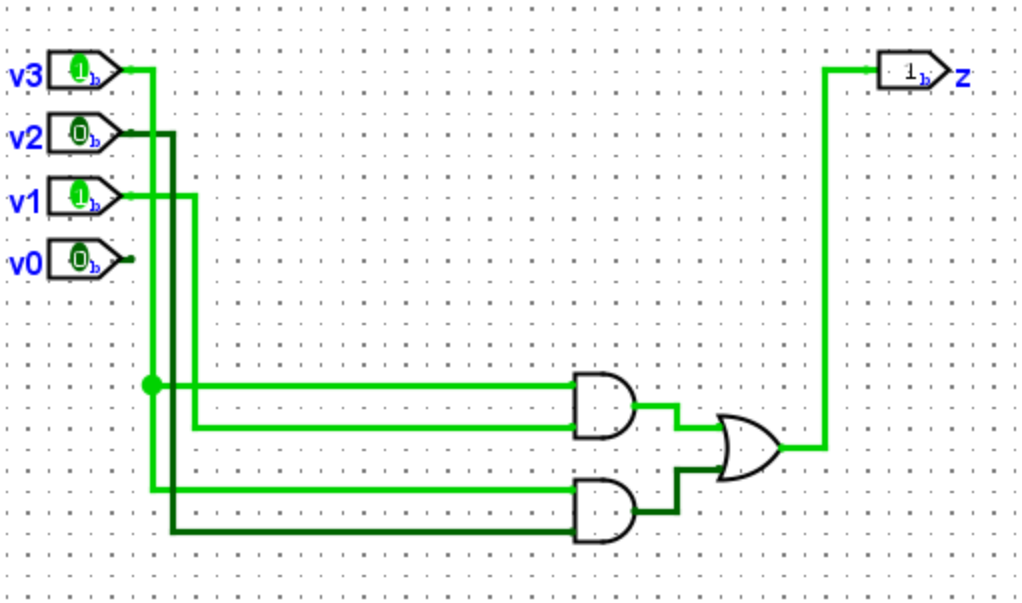
Truth Table:

i_m3	i_m2	i_m1	i_m0	o_seg[7..0]
0	0	0	0	1 1 0 0 0 0 0 0
0	0	0	1	1 1 1 1 1 0 0 1
0	0	1	0	1 0 1 0 0 1 0 0
0	0	1	1	1 0 1 1 0 0 0 0
0	1	0	0	1 0 0 1 1 0 0 1
0	1	0	1	1 0 0 1 0 0 1 0
0	1	1	0	1 0 0 0 0 0 1 1
0	1	1	1	1 1 1 1 1 0 0 0
1	0	0	0	1 0 0 0 0 0 0 0
1	0	0	1	1 0 0 1 1 0 0 0
1	0	1	0	1 1 0 0 0 0 0 0
1	0	1	1	1 1 1 1 1 0 0 1
1	1	0	0	1 0 1 0 0 1 0 0
1	1	0	1	1 0 1 1 0 0 0 0
1	1	1	0	1 0 0 1 1 0 0 1
1	1	1	1	1 0 0 1 0 0 1 0

myComparator:

Circuit that determines if there
is overflow into decimal

Circuit:

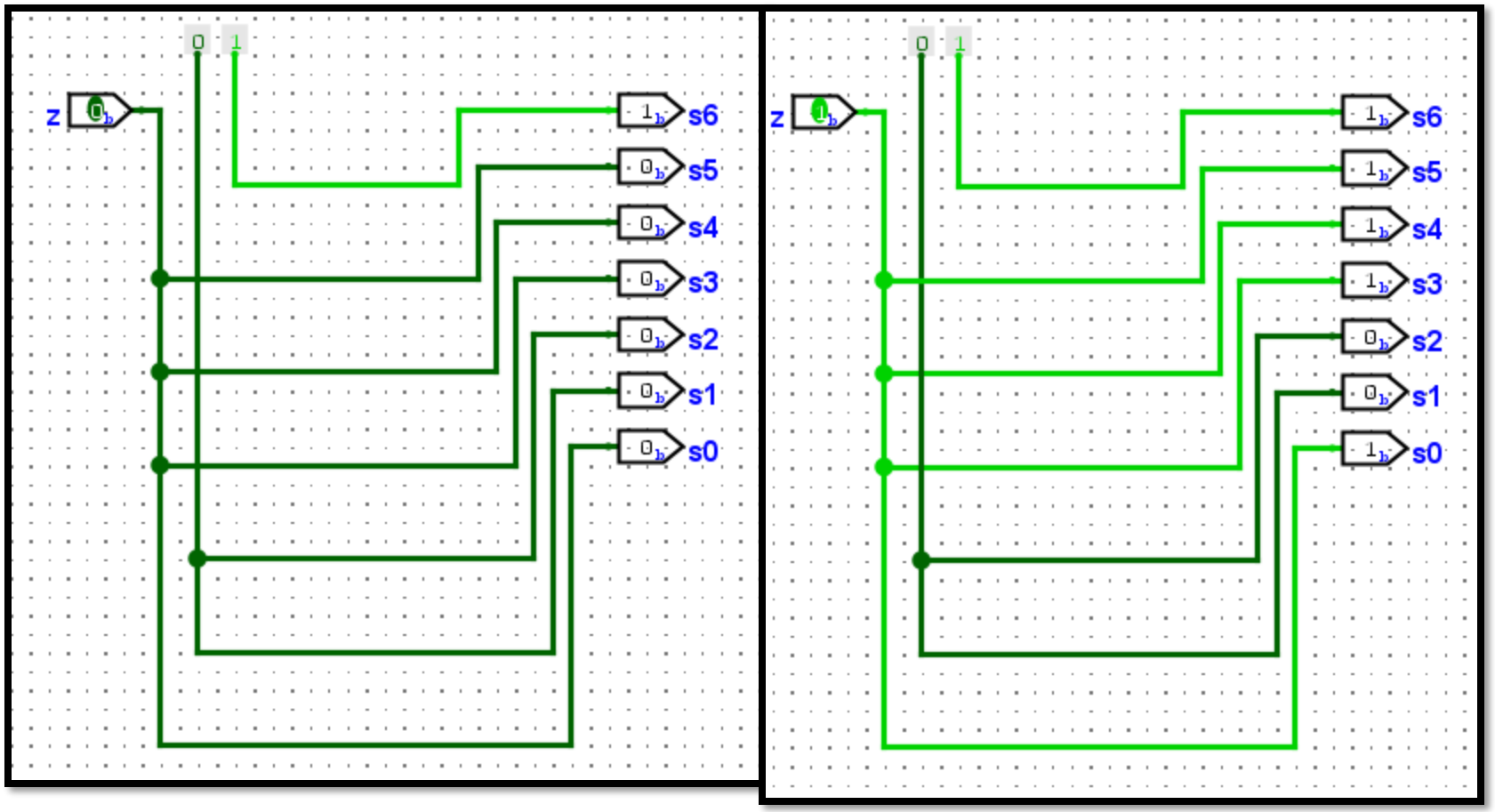


Truth Table:

v3	v2	v1	v0	z
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Circuit B: States whether decimal 10 is 1 or 0

Circuit:



Truth Table:

z	s6	s5	s4	s3	s2	s1	s0
0	1	0	0	0	0	0	0
1	1	1	1	1	0	0	1